

Practical PCB Design

ECEN 3730

Board 3: Golden Arduino

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Project Overview:

This board was designed to be a fully functional Arduino based on the commercial Uno R3 boards. Goals for this board were to learn how to design a complex, two-layer microcontroller board, and implement features to improve noise figures (power rail coupling, near-field emissions, cross talk, etc.) compared with a commercial board. The schematic design and board layout of my Golden Arduino referenced designs by Tim Swettlen, [SparkFun](#), and [Ruthvik Chanda](#).

Circuit Plan and Theory:

This board should have the following design features:

- 5V power from EITHER wall wart OR USB port
- ATmega328 bootload-able into Arduino
- Be able to run any Arduino sketch
- Be able to interface with standard Arduino shields
- Have near field emissions less than 10% of those of a commercial board

Figure 1 below shows a preliminary block diagram of the Golden Arduino. Figure 2 shows the schematic built around the block diagram in Figure 1.

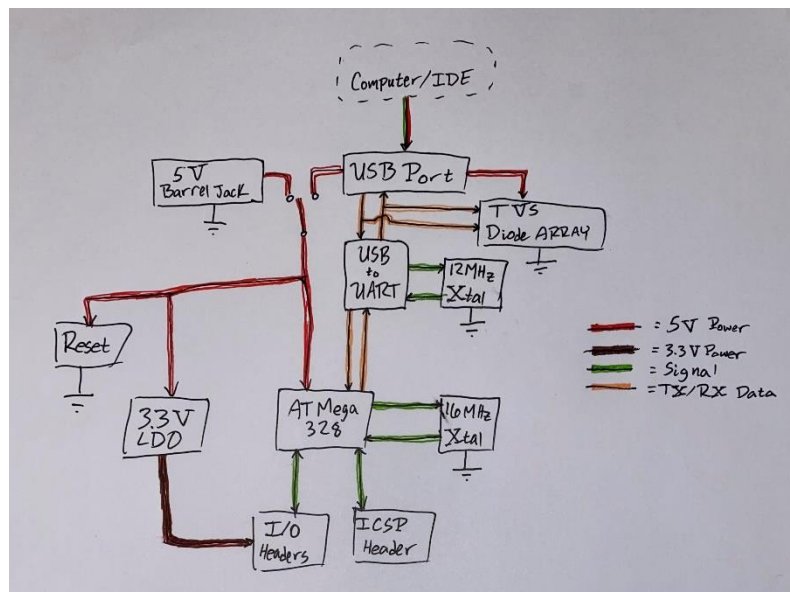


Fig. 1 (Golden Arduino block diagram)

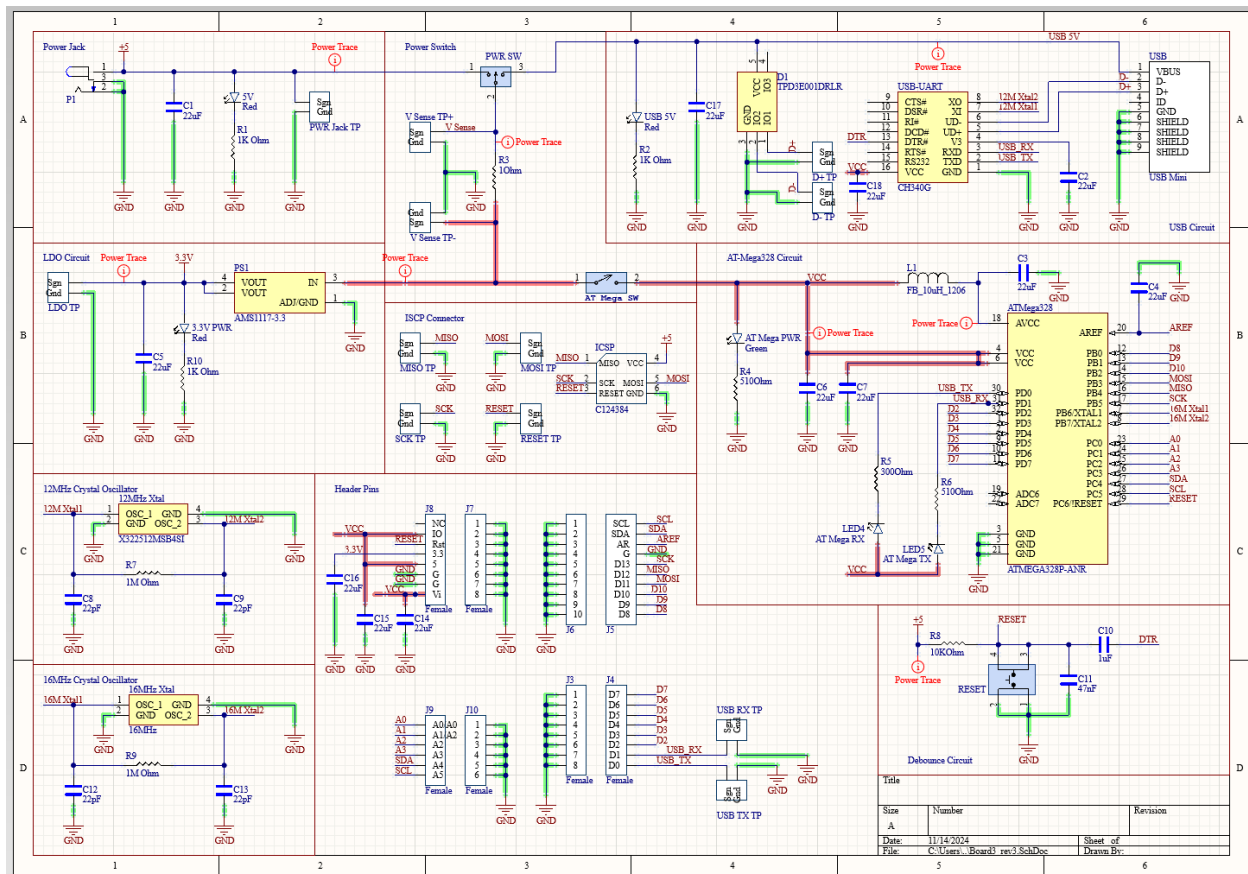


Fig. 2 (Schematic of Golden Arduino)

Design features implemented:

- 3-way header switch added to isolate barrel jack power from USB power
- Isolation switch from power to ATmega
- R3 allows for differential measurement of the current drawn by the ATmega
- Ferrite filter added to AVCC line to reduce switching noise at analog pin
- TVS diode array in USB circuit protects USB data lines from voltages exceeding USB power
- “USB_TX” net connects USB transmit line to ATmega receive line
- “USB_RX” net connects USB receive line to ATmega transmit line
- Test points on data lines to check bus data

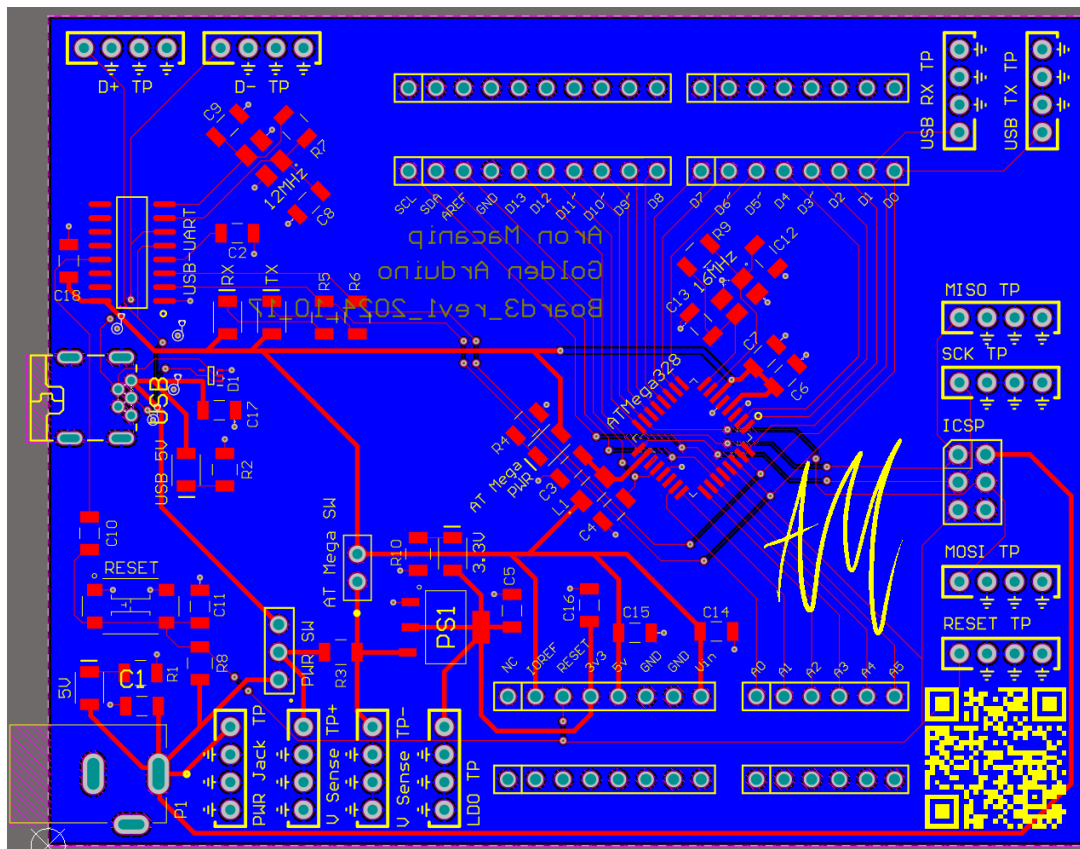


Fig. 3 (PCB layout of Golden Arduino)

- Headers spaced to match commercial Arduino Uno boards for compatibility with shields
- Second row of grounded headers parallel to signal headers allow local ground connections and shorter signal return paths for external components/breakouts
- Decoupling capacitors placed as close as practical to associated power pins
- Crystal oscillator circuits kept close to associated ICs to minimize loop inductance and potential noise coupling at IC oscillator pins
- ATmega power line routed through ground plane to minimize signal trace cross-unders
- All necessary cross-unders kept as short as possible
- Test points placed along outer edges for easy access

Physical Board:

The physical board is shown in Figures 4a and 4b below.

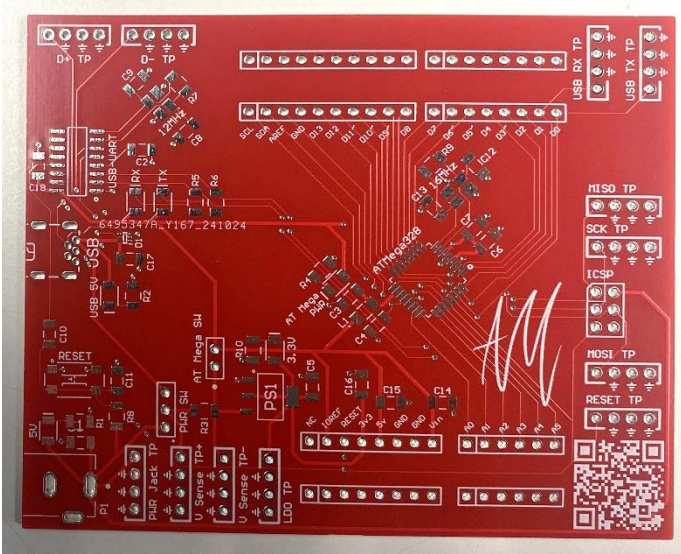


Fig. 4a (Unpopulated board)

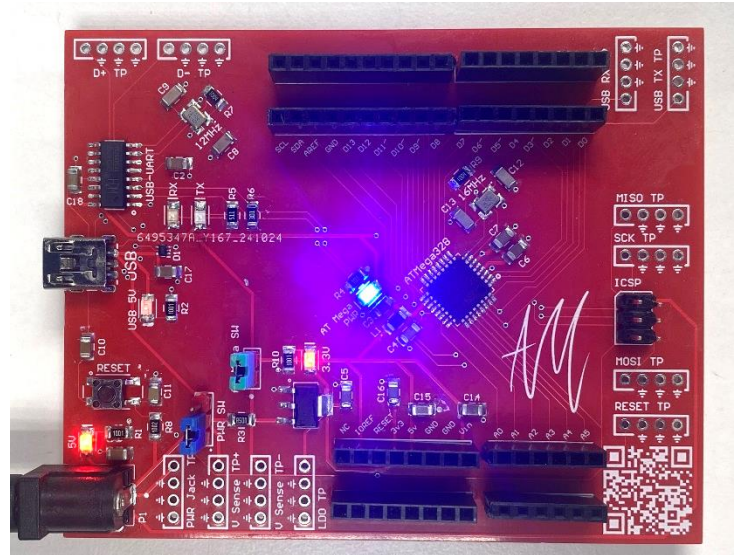


Fig. 4b (Fully populated board powered by barrel jack)

- C10 changed to 22 μ F, C11 changed to 1 μ F for better switch debounce
- R3 changed to 1.5 Ω due to hardware availability
- LED colors and associated current limiting resistors designed for changed due to available hardware
- Small leakage from barrel jack power back to USB port
 - Soft error, no significant effect on board operation
 - USB indicator LED on without USB plugged in, hard to see in Figure 4b, very faint
 - CH340G receives power anytime ATmega receives power (see “USB Circuit” block of schematic in Figure 2)

Troubleshooting and Debugging

Power, USB, and LDO circuits were populated and powered on independently to verify 5V output and reset function for the power and USB, and 3.3V output for the LDO. Remaining parts were populated and attempted to bootload, but bootload was unsuccessful.

- Signature mismatch error (0x000000 and 0x00ff00 signature), expected pin mismatch or short somewhere
- All ATmega pins checked against each other for continuity, no traces or pins misrouted
- 2 SPI (most likely MISO, MOSI) pins found to be shorted to ground under ATmega IC
- ATmega removed and replaced, pads cleaned, much less solder paste applied
- Continuity checked before attempting bootload

After spending 11 hours checking all other possible potential error points, a simple continuity check on all pins (taking all of 5 minutes) revealed a simple fix. Bootload was successful after resoldering.

Code to blink pin D13 at 5 and 50Hz was uploaded to verify board's functionality as an Arduino. Output is shown in Figures 5a and 5b below.

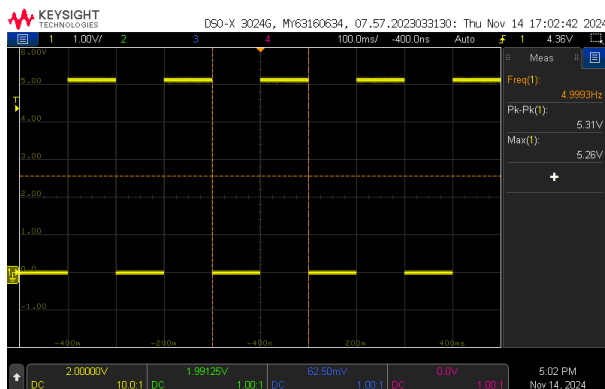


Fig. 5a (5Hz D13 blink)

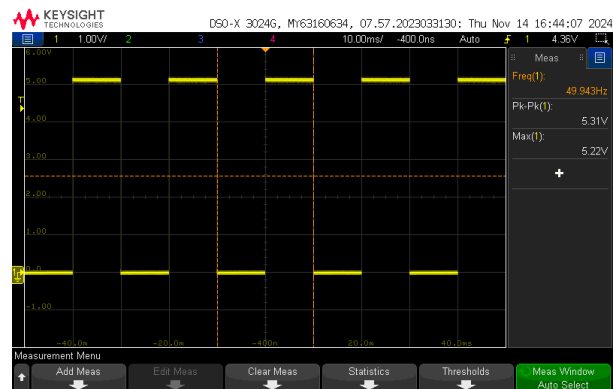


Fig. 5b (50Hz D13 blink)

In-Rush and Average Current

In-rush and average current pulled by ATmega circuit measured differentially across R3. Values are taken as pink oscilloscope traces shown in Figure 6 and Figure 7 respectively. Measurements and associated calculations are summarized in Table 1 below.

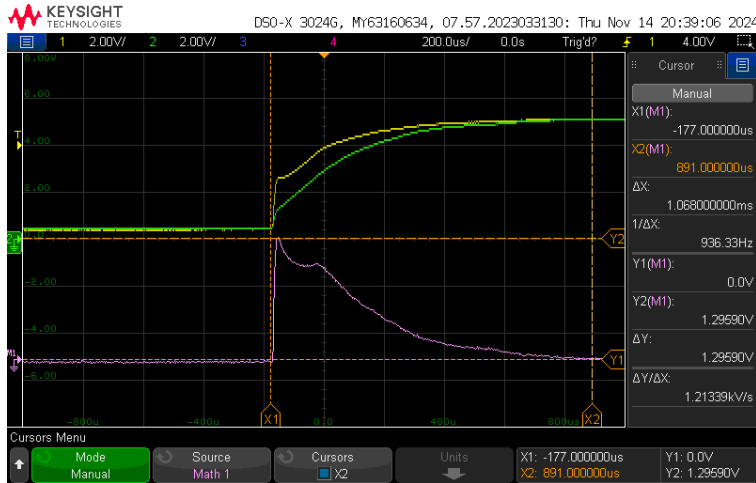


Fig. 6 (In-rush current measurement)

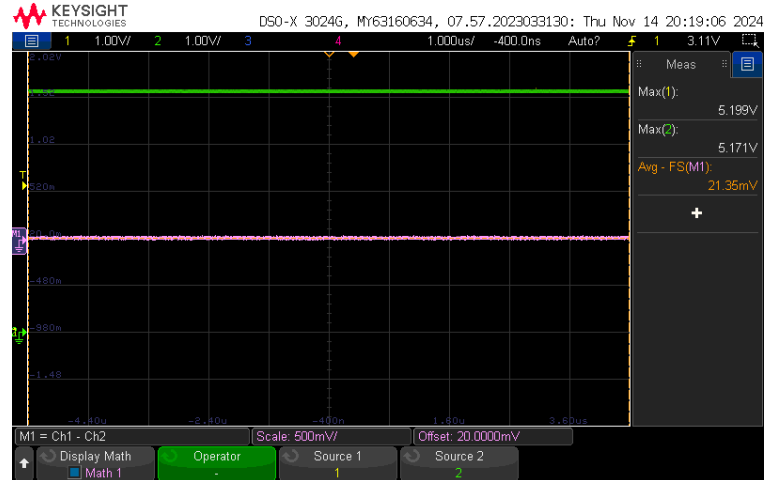


Fig. 7 (Average current measurement)

In-Rush Current and Power Calculations

$$\frac{1.3V}{1.5\Omega} \approx 0.867A \quad \frac{(1.3V)^2}{1.5\Omega} \approx 1.13W \quad 1.13W * 1.068ms \approx 1.2mJ$$

Average Current and Power Calculations

$$\frac{21.4mV}{1.5\Omega} \approx 14.3mA \quad \frac{(21.4mV)^2}{1.5\Omega} \approx 0.305mW$$

	Voltage	Current Draw	Power Dissipated	Energy Delivered
In-Rush	1.3 V	867 mA	1.13 W	1.2 mJ
Average	21.4 mV	14.3 mA	0.305 mW	

Table 1 (In-rush and average current measurement summary)

- In-rush power of ~1W dissipates over ~1ms
- Associated temperature change will be on the order of milli-Kelvin, staying well within ATmega's temperature rating
- Average current draw on power rail << 200mA rating for ATmega V_{CC} pin

Comparison to Commercial Board

Near-Field Emissions

Setup for near-field emission measurements using Golden Arduino is shown in Figure 8. Setup using commercial board is the similar. Figures 9a and 9b show measurements from the Golden Arduino and commercial board respectively.

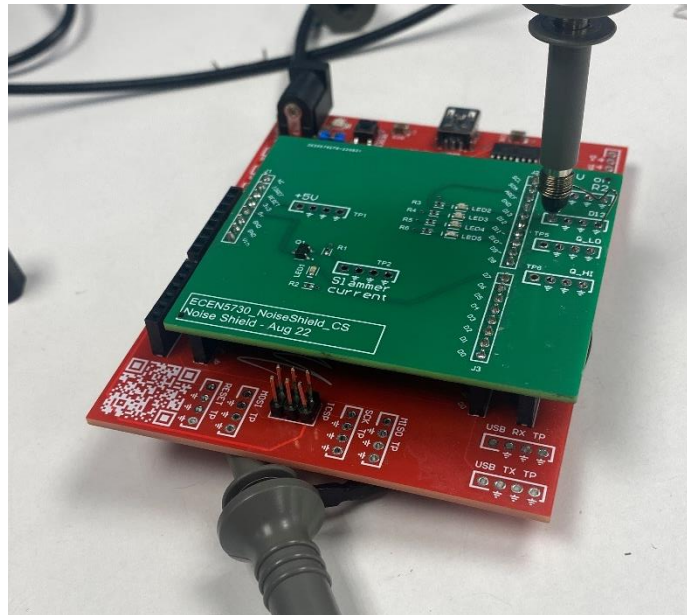


Fig. 8 (Near-field emission measurement setup)

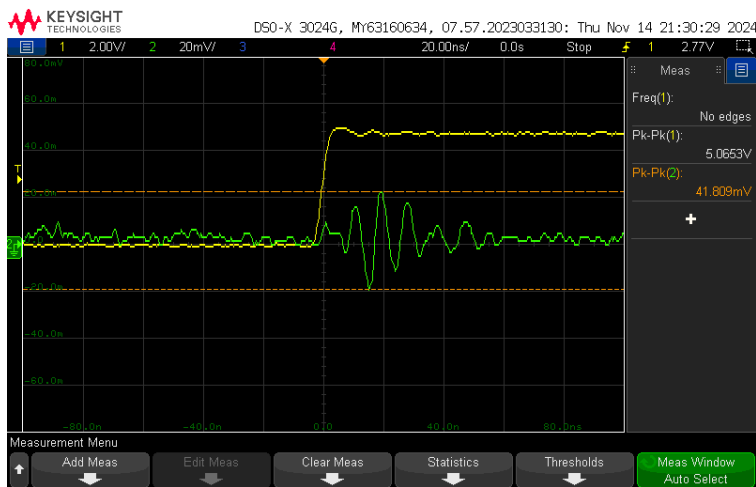


Fig. 9a (Golden Arduino near-field emissions)



Fig. 9b (Commercial board near-field emissions)

Switching noise on each line listed is shown in the Figures below. Golden Arduino measurements are on the left, commercial board measurements on the right, and rising edge measurements are above falling edge measurements where appropriate. All measurements are summarized in Table 2.

Power Rail

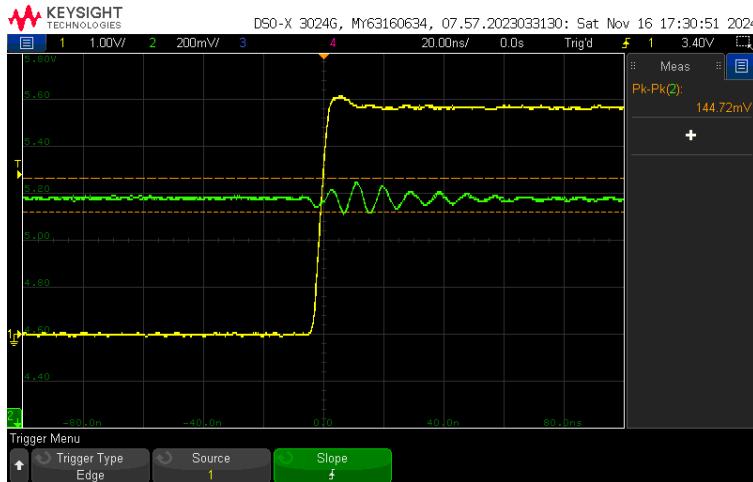


Fig. 10a (Golden Arduino power rail rise)

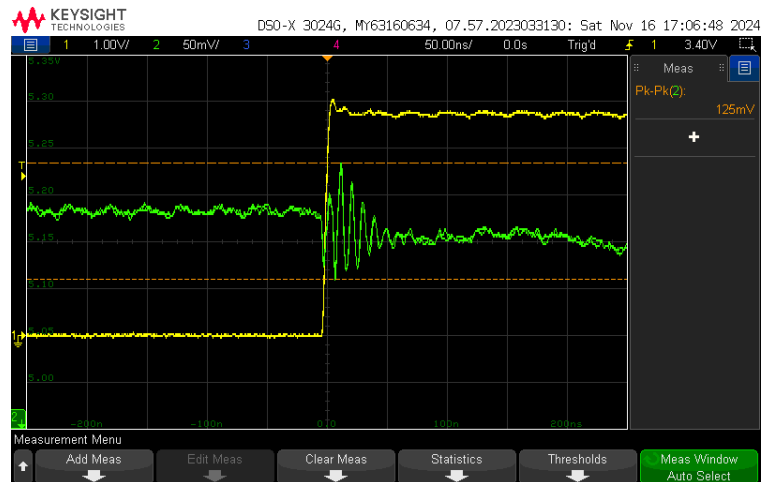


Fig. 10b (Commercial board power rail rise)

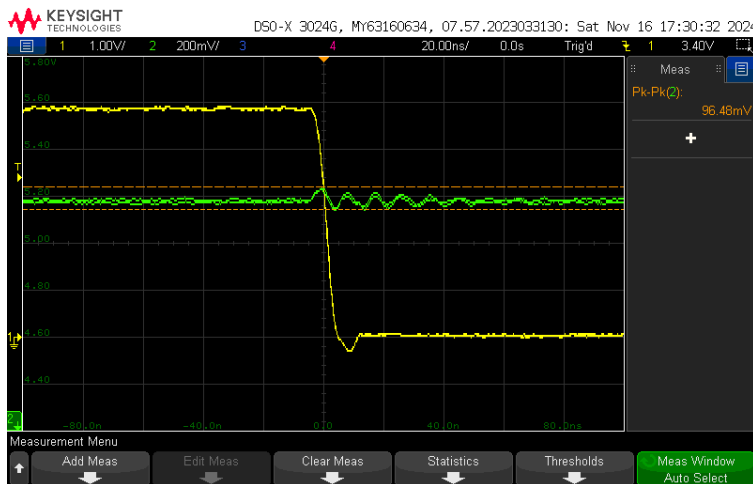


Fig. 11a (Golden Arduino power rail fall)

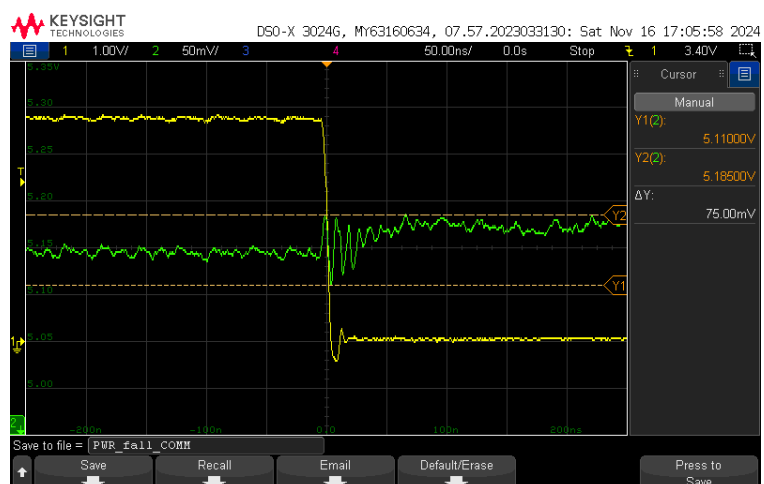
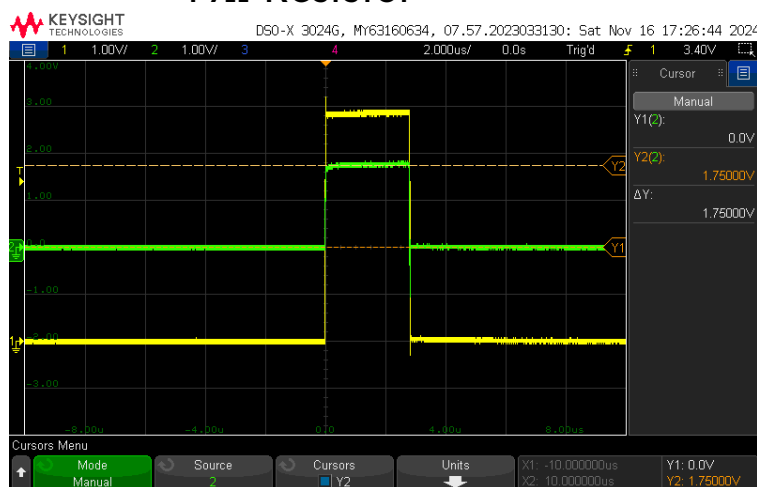
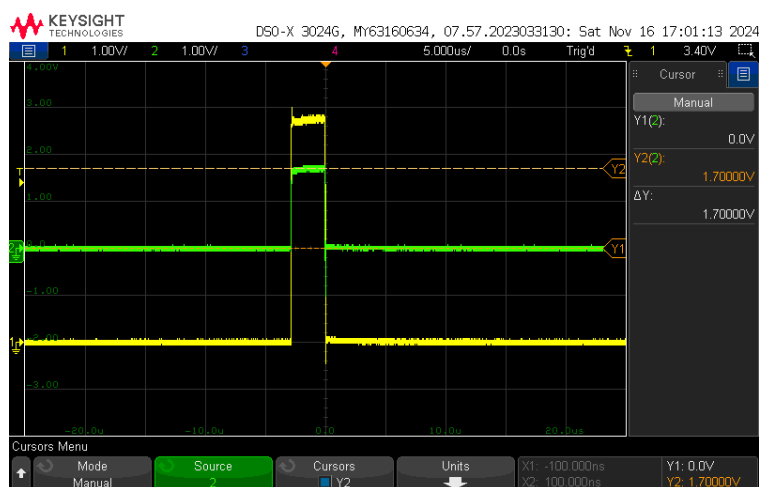


Fig. 11b (Commercial board power rail fall)

49 Ω ResistorFig. 12a (Golden Arduino 49 Ω resistor current)Fig. 12b (Commercial board 49 Ω resistor current)

Quiet High

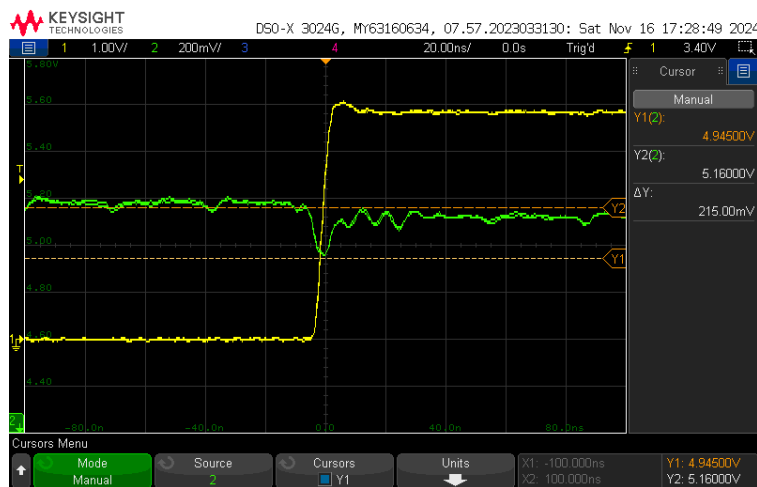


Fig. 13a (Golden Arduino quiet high rise)

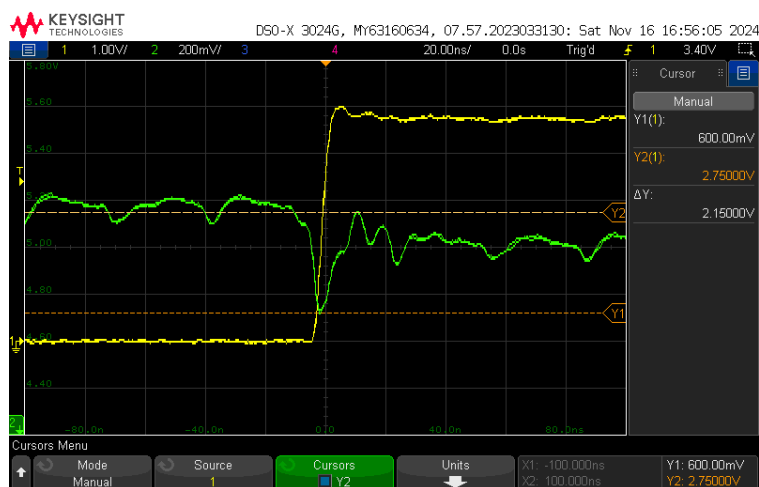


Fig. 13b (Commercial board quiet high rise)

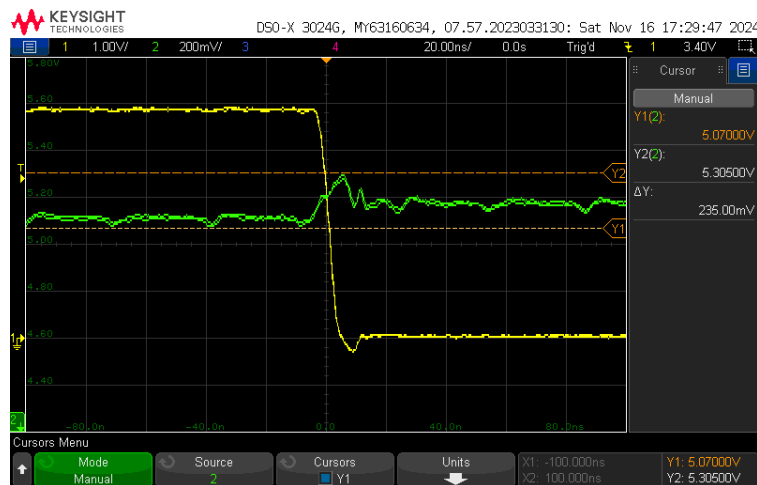


Fig. 14a (Golden Arduino quiet high fall)

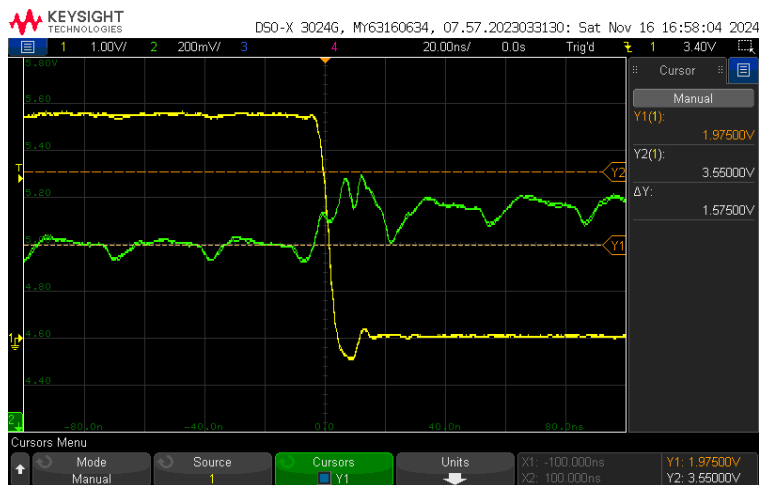


Fig. 14b (Commercial board quiet high fall)

Quiet Low

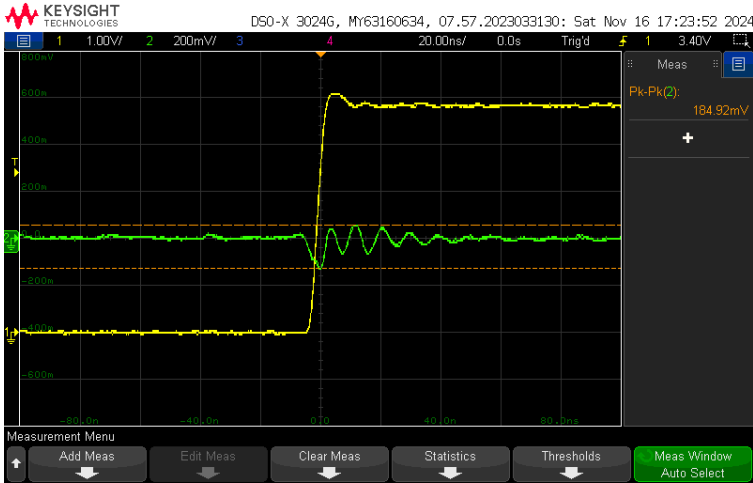


Fig. 15a (Golden Arduino quiet low rise)

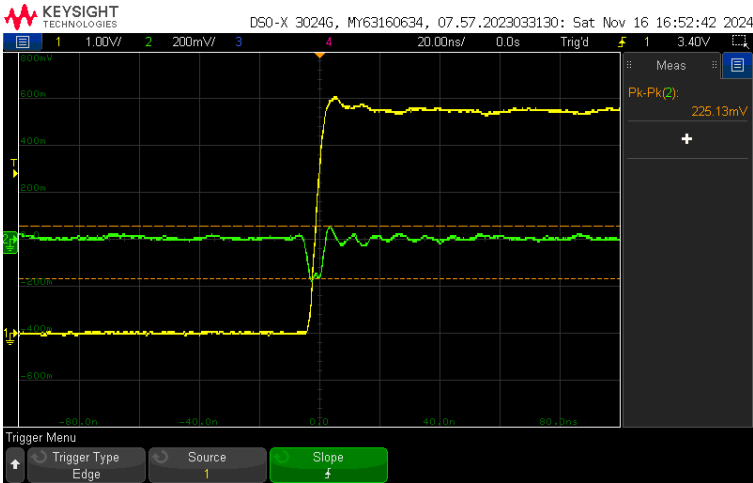


Fig. 15b (Commercial board quiet low rise)

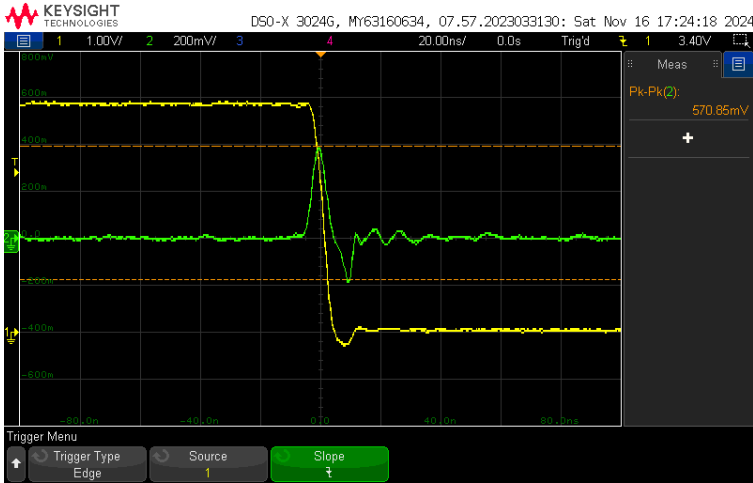


Fig. 16a (Golden Arduino quiet low fall)

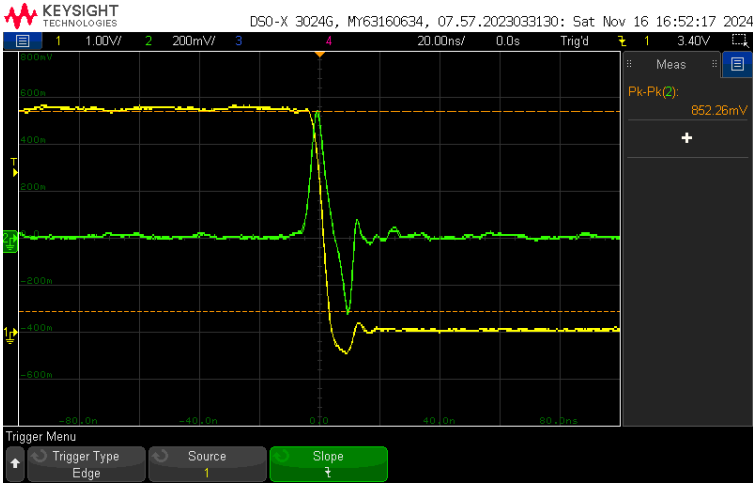


Fig. 16b (Commercial board quiet low fall)

Slammer Circuit

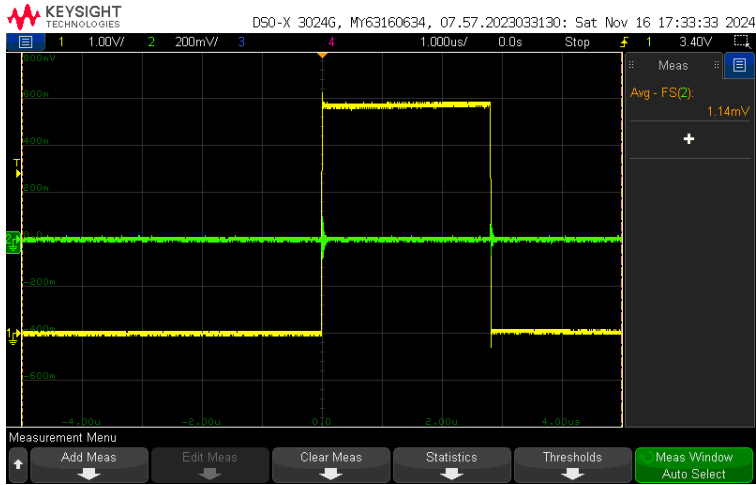


Fig. 17a (Golden Arduino slammer average)

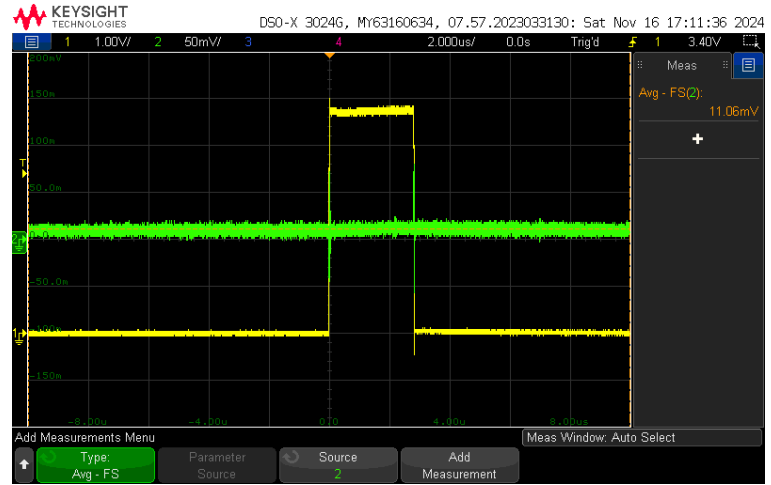


Fig. 17b (Commercial board slammer average)

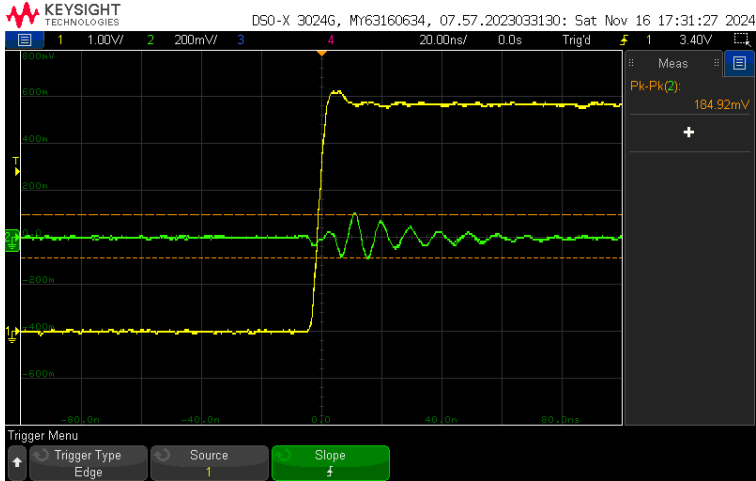


Fig. 18a (Golden Arduino slammer rise)

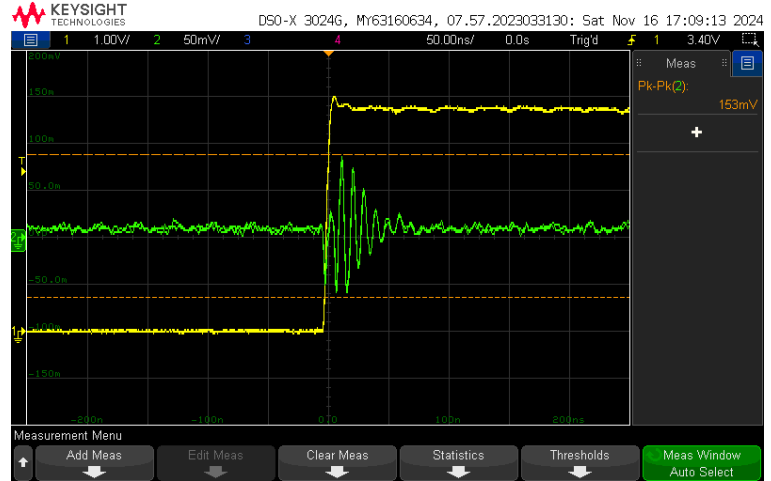


Fig. 18b (Commercial board slammer rise)

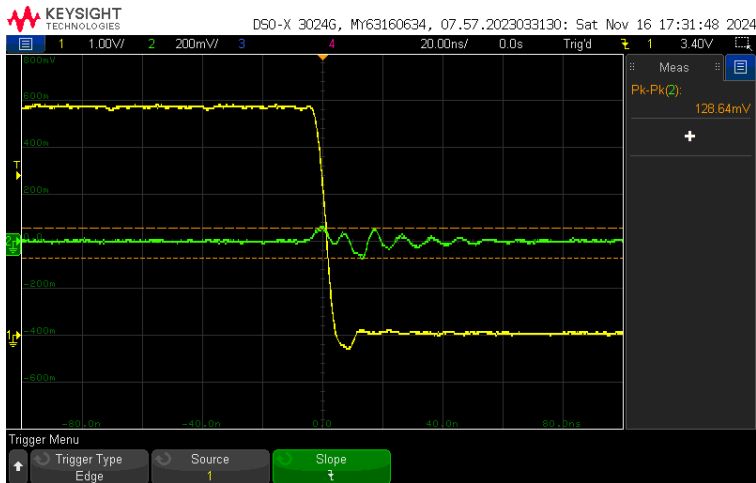


Fig. 19a (Golden Arduino slammer fall)

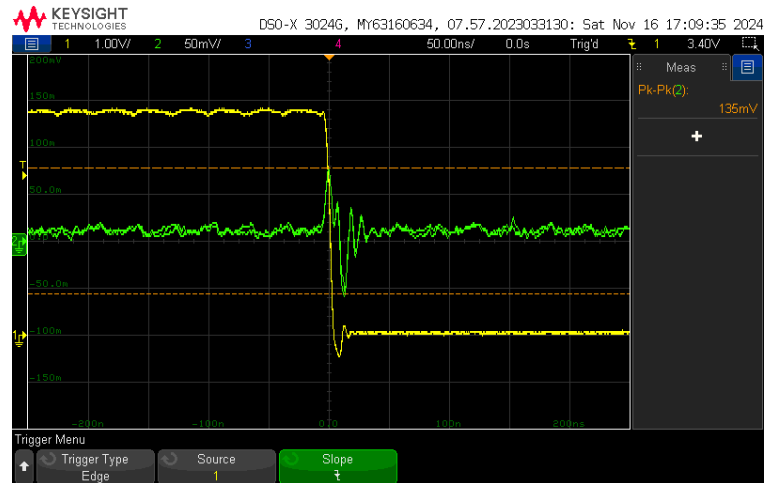


Fig. 19b (Commercial board slammer fall)

Measurement Summary

	Golden Arduino	Commercial Board
Near-Field	41.8 mV Pk-Pk	93.3 mV Pk-Pk
Power (Rise)	0.145 V Pk-Pk	0.125 V Pk-Pk
Power (Fall)	96.5 mV Pk-Pk	75mV Pk-Pk
Quiet High (Rise)	0.215 V Pk-Pk	2.15 V Pk-Pk
Quiet High (Fall)	0.235 V Pk-Pk	1.58 V Pk-Pk
Quiet Low (Rise)	0.571 V Pk-Pk	0.225 V Pk-Pk
Quiet Low (Fall)	0.185 V Pk-Pk	0.852 mV Pk-Pk
Slammer (Avg)	1.14 mV	11.1 mV
Slammer (Rise)	0.185 V Pk-Pk	0.153 V Pk-Pk
Slammer (Fall)	0.129 V Pk-Pk	0.135 V Pk-Pk
49Ω (Voltage)	1.75 V	1.7 V
49Ω (Current)	35.7 mA	34.7 mA

Table 2 (Golden Arduino and commercial board noise figure summary)

- Commercial board near-field emissions were ~35mV peak-to-peak at 33.3MHz with steady signal
- Golden Arduino has ~6mV peak-to-peak near-field emissions with no discernable periodic at steady signal, ~16.7% of commercial board
- Boards push comparable currents to 49Ω resistor
- Golden Arduino peak-to-peak near-field emissions on digital switching is ~45% of the commercial board emissions on digital switching
- Commercial board has slightly better power rail noise suppression
 - Most likely culprit is placement of ferrite filter and AVCC decoupling capacitor
 - Ferrite placed closer to AVCC pin; loop inductance is higher than practical min
- Average slammer noise of Golden Arduino ~10% of commercial board
- Commercial board has slightly better rise noise figures, but much worse fall noise figures

Areas for Improvement

- Add on-board LED to D13 for blinking
- Add crystal test points to more easily check for oscillations after bootload
- Avoid placing signal vias under ATmega to prevent shorting to ground when soldering
- Change CH340G VCC pin to “USB 5V” net instead of “VCC” net
- Swap placement of C3 and L1 to reduce loop inductance to ATmega’s AVCC pin
- Space signal cross-unders to minimize ground plane peninsulas formed (specifically under ATmega)

Conclusions:

- Avoid placing signal vias under components as much as practical
- Check for proper continuity after assembly before any other test
- Use less solder paste, especially if there are vias underneath IC
- Ground pour on top signal layer can increase coupling noise and/or near-field emissions for digital data transfer frequencies